# XAIMed-Net: towards explainable brain tumour detection in 2D T1-weighted CE-MRI images using transfer learning

**Conference Paper** *in* IET Conference Proceedings · September 2024

DOI: 10.1049/icp.2024.3305



**5 authors**, including:

Vayangi Ganepola
Dublin City University
**6** PUBLICATIONS **14** CITATIONS

Julia Dietlmeier
Dublin City University
**28** PUBLICATIONS **128** CITATIONS

Claudia Mazo
Dublin City University
**54** PUBLICATIONS **889** CITATIONS

Noel O'Connor
Dublin City University
**652** PUBLICATIONS **11,574** CITATIONS

# XAIMed-Net: Towards Explainable Brain Tumour Detection in 2D T1-weighted CE-MRI Images Using Transfer Learning

Oluwabukola Adegboro[1], Vayangi Ganepola[1], Julia Dietlmeier[2], Claudia Mazo[3], Noel E. O'Connor[2]

[1] *SFI Centre for Research Training in Machine Learning, Dublin City University, Ireland*
[2] *Insight SFI Research Centre for Data Analytics, Dublin City University, Ireland*
[3] *School of Computing, Dublin City University, Ireland*

**Abstract**

Malignant brain tumours comprise one of the deadliest types of cancer and their timely and accurate detection is receiving urgent attention. Radiology can benefit from the recent impressive progress of AI and deep learning that provide the non-invasive diagnostic opportunity. However, the challenge of explainable AI-based decision-making processes remains. In this work, we aim to take a step toward more explainable classification of brain tumours in medical images by designing a new convolutional architecture while also leveraging the advantages of transfer learning. Specifically, we introduce a new shallow convolutional head that can be added to a pre-trained ResNet50 model to produce EigenCAM-based importance heatmaps. We investigate pre-training based on the ImageNet and RadImageNet datasets. Furthermore, we also address the problem of weakly-supervised brain tumour segmentation in a challenging 2D T1-weighted CE-MRI dataset by developing an approach to post-process the resulting heatmaps. Our preliminary experimental results with the Dice Score Coefficient of 0.145 and the new objective distance metric $\mu\Delta = 59.55$ show the potential of combining large pre-trained models with shallow convolutional architectures to localize tumours using only weak classification labels. Our code is publicly available at https://github.com/juliadietlmeier/Explainable_brain_tumor_detection_in_MRI.

**Keywords:** Brain tumours, EigenCAM, Explainable Artificial Intelligence, Transfer learning, Weakly-Supervised Segmentation.

## 1 Introduction

A brain tumour is a lump-like foreign body that occurs in the human brain due to the growth of abnormal cells. Every year, 450 people in Ireland are diagnosed with primary brain tumours[1] originating from the brain or its surroundings. Brain tumour symptoms include headaches, vision and speech problems[2]. The survival rate depends on the early diagnosis. Hence effective brain tumour detection techniques, including the use of Explainable AI (XAI) are needed. XAI helps users understand the output of an AI model and the inner process taken to achieve that outcome. It is also referred to as a "White box" because it describes or explains the model process [Saranya A., 2023]. Also, it is commonly associated with terms such as Transparency in Machine Learning, Interpretability etc. [Bernal and Mazo, 2022, Antoniadi et al., 2021]. Within the healthcare domain, some XAI techniques proposed to date include; SHAP (SHapley Additive exPlanations), CAM (Class Activation Maps), and LIME [Saranya A., 2023]. In this paper, we apply an XAI technique called EigenCAM [Muhammad and Yeasin, 2020] for the identification and Weakly-Supervised Segmentation (WSS) of brain tumours. Our major contributions are as follows:

[1] https://www.cancer.ie/cancer-information-and-support/cancer-types/brain-tumours
[2] https://www.nhs.uk/conditions/brain-tumours/

https://doi.org/10.1049/icp.2024.3305

- First, we address the task of explainable brain tumour detection in Magnetic Resonance Imaging (MRI) data using transfer learning and propose a new convolutional head, based on the context of a SqueezeNet [Iandola et al., 2017] architecture. This head is added to a pre-trained ResNet50 [He et al., 2015] backbone, which was initially trained on large-scale RadImageNet [Mei et al., 2022] and ImageNet [Russakovsky et al., 2015] datasets.
- Second, we develop a new post-processing pipeline to threshold the importance heatmaps returned by EigenCAM, thus addressing the task of weakly-supervised segmentation of brain tumours.

# 2 State of the Art

Over the past years some open-source MRI datasets useful for identifying brain tumours have been proposed. Some state-of-the-art techniques applied on these datasets are summarised as follows. In [Chatterjee et al., 2023], three inherently-explainable classifiers are proposed; GP-UNet, GP-ReconResNet, and GP-ShuffleUNet for the multi-class brain tumour classification and weakly-supervised segmentation task and the Jun Cheng dataset [Cheng et al., 2015] was used for evaluating the model. When compared to other classifers such as InceptionV3 and ResNeXt50, the top performing model for the Jun Cheng dataset is GP-ReconResNet. GP-ReconResNet achieved an F1-score of 0.93, and a median Dice score of $0.10 \pm 0.0365$ for the classification task and weakly-supervised segmentation task, respectively. On the other hand, the GP-UNet obtained an F1-score of 0.93 for classification, and a median Dice score of $0.67 \pm 0.08$ for weakly-supervised segmentation task.

On the other hand, [Ahmed et al., 2023] introduced a VGG16-based model architecture coupled with an XAI technique known as LRP (Layer-wise Relevance Propagation) for identifying and localizing brain tumours. While the VGG16 model was used to classify if the MRI brain images were 'Normal' or 'Tumour', the LRP on the other hand is used to localize the tumour regions in a brain MRI scan, hence providing more insights on the decision-making process of whether a tumour exists or not. The dataset was trained on the Br35H dataset [Hamada, 2020]. The results of the proposed model outperformed other prior related works with a testing accuracy of 97.33%, a misclassification rate of 2.67% and an F1-score of 0.974.

To summarize, we found only a limited amount of papers ([Chatterjee et al., 2023], [Schiavon et al., 2023], [Gaur et al., 2022], [Zeineldin et al., 2022], [Kumar et al., 2021]) which developed explainable networks for the localization and weakly-supervised segmentation of MRI brain tumour datasets. As we show in our experimental section, our quantitative WSS results surpass the results reported in [Chatterjee et al., 2023].

# 3 Methodology

In this work we design an explainable classification model termed XAIMed-Net to classify brain tumours (meningioma, glioma, pituitary tumour) in the 2D T1-weighted CE-MRI dataset [Cheng et al., 2015].

## 3.1 Proposed XAIMed-Net architecture

The novelty of our approach is that we introduce a new Convolutional Head (CH) that we add to the pre-trained ResNet50 backbone. For this, we loosely follow the idea of SqueezeNet [Iandola et al., 2017] which employs alternating *squeeze* and *expansion* blocks. A squeeze block in our case employs $1 \times 1$ convolutions and performs dimensionality reduction by a factor of $r$. In contrast to the *Fire* module of SqueezeNet, our expansion block increases the number of channels by a factor of $r$ using only $3 \times 3$ convolutions. Our proposed explainable classification pipeline is shown in Figure 1.

We empirically observe that XAIMed-Net models pre-trained on either ImageNet or on RadImageNet do not overfit in the first 100 epochs, so there is no need to add Dropout layers. The last convolutional layer (in the 3rd expansion block) is used to obtain EigenCAM [Muhammad and Yeasin, 2020]. We leverage transfer learning via pre-trained ResNet50 model and remove the last convolutional stage to increase the output resolution of the heatmaps to be $14 \times 14$ pixels. During training, we freeze the remaining ResNet50 stages and finetune the

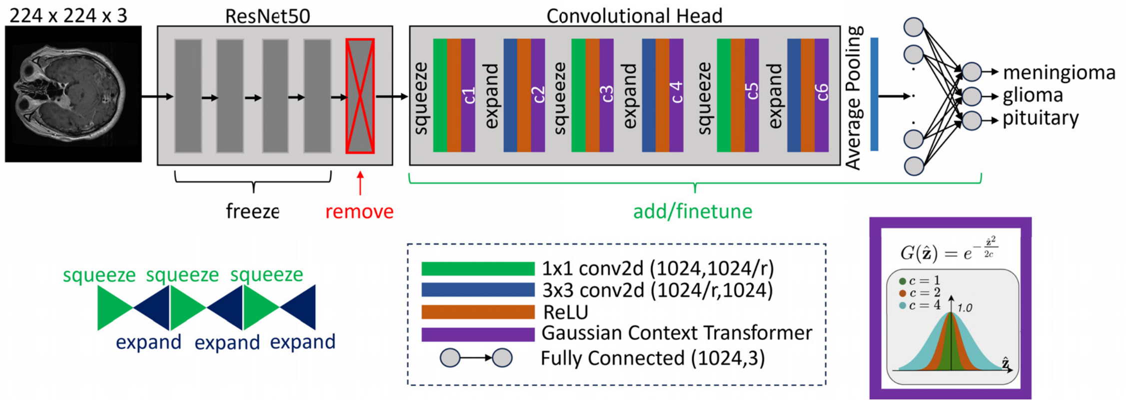


Figure 1: Block diagram of the proposed classification network XAIMed-Net for explainable brain tumour detection on the [Cheng et al., 2015] dataset. In a nutshell, our XAIMed-Net consists of a pre-trained ResNet50-based backbone and a new convolutional head with three squeeze-and-expand blocks. To increase the output resolution to $14 \times 14$ we remove the last stage of ResNet50 and freeze the remaining stages during training.

convolutional head + average pooling layer + classifier. The squeeze block performs dimensionality reduction by a factor of $r$ (with a $1 \times 1$ convolutional kernel). Each squeeze and expansion block consists of a convolutional layer, ReLU (Rectified Linear Unit) activation function and a Gaussian Context Transformer (GCT) [Ruan et al., 2021] attention layer. The inclusion of GCTs has an important impact on the heatmap's shape and improves classification accuracy.

We follow a systematic methodology to obtain the optimal $r$ parameters. This set of experiments is shown in Figure 3. For the XAIMed-Net pre-trained on ImageNet, we empirically obtain the best possible configuration for $r = 2$. Next, we empirically obtain the Gaussian spread parameter $c = 1$ for the fixed $r = 2$. Finally, we obtain $r = 1$ and $c = 1$ configurations for the XAIMed-Net pre-trained on RadImageNet in the same way.

To explain classification decisions made by XAIMed-Net, there is a number of CAM-based methods available. Although they all have their merits and may produce different explanations for the same model, we argue that the resulting heatmap primarily depends on the architecture of a classification network. We select the EigenCAM [Muhammad and Yeasin, 2020] method as it is claimed to produce particularly focused heatmaps, which we found to be easily thresholded using our post-processing approach outlined in Figure 2.

Furthermore, we attempt to solve weakly-supervised segmentation of brain tumours by using the thresholded heatmaps returned by the selected EigenCAM explainability algorithm. To process the resulting heatmaps we develop a post-processing pipeline illustrated in Figure 2.

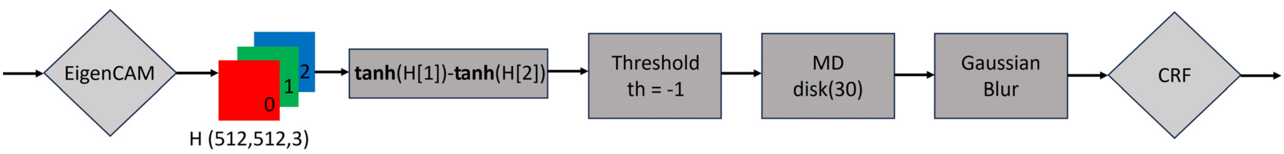


Figure 2: Proposed WSS post-processing pipeline to obtain the predicted mask $\hat{M}$. MD is the morphological dilation operation with a disk of size 30. We specifically include the difference of $tanh$ based operation to effectively extract the core (area of highest importance) of the predicted heatmap $H$.

Specifically, to increase segmentation precision we employ Conditional Random Fields (CRF) to refine the thresholded localization heatmaps. CRF is a probabilistic graphical model of maximum a posteriori with Gibbs distribution [Sutton and McCallum, 2010]. The proposed post-processing pipeline operates on the $512 \times 512$ resolution of RGB heatmaps $H$. Initially, we find it difficult to select an optimal threshold to threshold $H$ directly. Therefore, we introduce a difference of $tanh$ between the first and second channel of $H$. The resulting modified heatmap can be easily thresholded with a threshold of $th = -1$.

# 4 Experiments

## 4.1 Datasets

The [Cheng et al., 2015] dataset[3] used in this study consists of 3,064 T1-weighted contrast-enhanced MRI image slices. This dataset was collected from 2005 to 2010 from Nanfang Hospital in Guangzhou, China, and General Hospital at Tianjing Medical University, China. Each slice information is represented in a `.mat` (MATLAB) format, containing the patient ID, image array, classification label, tumour border array, and tumour mask. The dataset was collected from 233 patients and the labels consist of three types of primary brain tumours. This includes: Meningioma (708 slices), Glioma (1,426 slices), and Pituitary tumour (930 slices). Samples are collected in three different orientations: axial, coronal, and sagittal. This large and imbalanced dataset is challenging due to variability on tumour image characteristics, shapes, and sizes. Finally, most images have an in-plane resolution of 512×512 pixels and some of 256×256 pixels.

From each tumour category we randomly extract 70%-10%-20% images for train-validation-test split and then combine the corresponding class folders for final training, validation and testing. This methodology is the same as the stratified random sampling. The final training folder consists of 2,146 images, validation folder has 308 images and test folder has 610 images. By using stratified random sampling we aim at preserving the original dataset distribution.

Pre-training datasets used in this work include the open source RadImageNet and ImageNet datasets. RadImageNet is a large database of annotated medical images from multiple modalities and of multiple pathologies. RadImageNet is comprised of five million labelled images of medical pathology with DICOM tags on over one million PET, CT, Ultrasound, and MRI studies from 500,000 patients [Mei et al., 2022]. RadImageNet PyTorch weights for the ResNet50 backbone are available[4]. ImageNet [Russakovsky et al., 2015] is a well-known large-scale dataset of more than one million natural images from 1000 different object classes.

## 4.2 Implementation and training details

The processing pipeline was implemented in Python 3.9 and the deep learning open source library PyTorch 2.0.1. All experiments were performed on a desktop computer with the Ubuntu operating system 18.04.3 LTS with the Intel(R) Core(TM) i9-9900K CPU, Nvidia GeForce RTX 2080 Ti GPU, and a total of 62GB RAM.

We finetune XAIMed-Net using an AdamW optimizer [Loshchilov and Hutter, 2019] with a learning rate of $lr = 0.00001$. Batch size is 16 and all models are trained for 100 epochs. During the XAIMed-Net training, we implement an early stopping mechanism (with patience=2, min_delta=0.5) conditioned on the training loss to prevent the training loss from large sudden changes in values. This is especially needed in the experiments (Figure 3) with ImageNet-based XAIMed-Net where we investigate extreme values of $r$. To ensure the reproducibility, all seeds are set to 42. As the dataset is slightly imbalanced, we use weighted Cross Entropy Loss as the main optimization objective where we compute the weights according to the number of images in each class. No data augmentation was used in the training phase. All images are resized to $224 \times 224$ as an input to XAIMed-Net. After inference we resize all heatmaps back to the original $512 \times 512$ resolution for post-processing and quantitative analysis. We apply standard PyTorch normalization as a pre-processing step.

## 4.3 Experimental Results

In this section, we present our preliminary results where we apply the XAIMed-Net to classify the Cheng et al. dataset. To measure how well the heatmaps focus on the tumour areas, we define an objective metric $\Delta$ to measure the distance between the ground truth segmentation mask $M$ and the predicted (see Figure 2) mask $\hat{M}$.

$$\Delta = ||p_1, p_2||_2 = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2} \tag{1}$$

[3] available on https://figshare.com /articles/dataset/brain_tumor_dataset/1512427

[4] https://github.com/BMEII-AI/RadImageNet

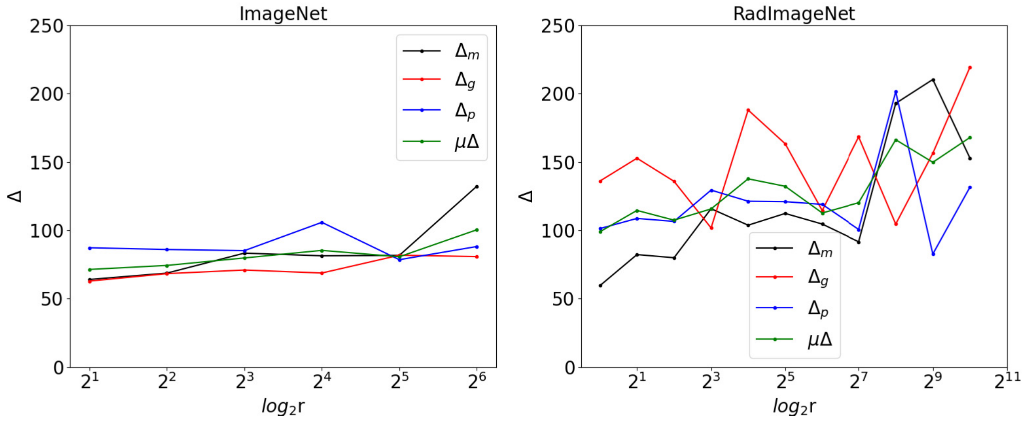


Figure 3: Finding out the value of the optimal reduction parameter $r$ (logarithmic x-axis). We vary $r$ from 1 to 1024 in steps of power of 2. We extract the optimal parameters $r$ at the minimum average distance $\mu\Delta = \mu(\Delta_m, \Delta_g, \Delta_p)$ between meningioma $\Delta_m$, glioma $\Delta_g$ and pituitary $\Delta_p$ distances $\Delta$. As can be seen from the curves: the optimal $r$(RadImageNet)=1 where $\mu\Delta_{min} = 98.96$. All $\Delta$ metrics are computed on the test set. Corresponding ImageNet and RadImageNet models (XAIMed-Net configuration without GCT layers) were trained for 100 epochs with the learning rate of $lr = 0.00001$. ImageNet-based extreme values of $\Delta$ for $r = (1, 128, 256, 512, 1024)$ are $NaN$. The optimal $r$(ImageNet)=2 where $\mu\Delta_{min} = 71.36$. It can be seen that ImageNet-based distances are lower than that of RadImageNet.

where $p_1 : \{x_1, y_1\}$ is the center of mass of the ground truth mask $M$ and $p_2 : \{x_2, y_2\}$ is the center of mass of the predicted mask $\hat{M}$. In addition, we also use the Dice Similarity Score (DSC), Jaccard coefficient and classification accuracy metrics, all computed on the test set. To obtain the optimal values of the reduction parameter $r$ we perform a series of experiments outlined in Figure 3. Quantitative results including an ablation study are shown in Table 1. Qualitative results are provided in Figure 4.

| Model | $\Delta_m \downarrow$ | $\Delta_g \downarrow$ | $\Delta_p \downarrow$ | $\mu\Delta \downarrow$ | Accuracy ↑ | WSS DSC ↑ | WSS Jaccard ↑ |
|---|---|---|---|---|---|---|---|
| SEA3 RadImageNet | **74.25** | **93.85** | **112.05** | **93.38** | 94.58% | 0.117 | 0.063 |
| SEA2 RadImageNet | 83.04 | 99.32 | 120.19 | 100.85 | 95.07% | **0.123** | **0.066** |
| SEA1 RadImageNet | 109.40 | 115.31 | 143.55 | 122.75 | **95.57%** | 0.096 | 0.051 |
| SEA0 RadImageNet | 119.92 | 142.85 | 139.64 | 134.14 | 91.63% | 0.052 | 0.027 |
| SEA3 ImageNet | 41.66 | **66.02** | **79.56** | 62.41 | 96.55% | 0.145 | **0.055** |
| SEA2 ImageNet | 33.43 | 69.47 | 118.73 | 73.88 | 93.92% | 0.145 | 0.042 |
| SEA1 ImageNet | **29.45** | 66.72 | 82.48 | **59.55** | **98.19%** | **0.145** | 0.052 |
| SEA0 ImageNet | 36.60 | 72.30 | 85.17 | 64.69 | 98.19% | 0.145 | 0.043 |

Table 1: Quantitative results and ablation study. SEA3 configuration refers to the full XAIMed-Net. SEA2 has only two squeeze-and-expand blocks. SEA1 has only one squeeze-and-expand block and SEA0 refers to the convolutional head with only one squeeze block. Optimal parameters for RadImageNet are $r = 1$ and $c = 1$. Optimal parameters for ImageNet are $r = 2$ and $c = 1$. Accuracy refers to the test classification accuracy.

As can be seen from the summary of quantitative results in Table 1, our XAIMed-Net in SEA1 ImageNet configuration with the WSS DSC of 0.145 outperforms the state-of-the-art GP-ReconResNet model with the reported WSS DSC of $0.10 \pm 0.0365$ [Chatterjee et al., 2023] while also having less number of model parameters (13.8M versus 17.3M). It can also be seen that ImageNet pre-trained SEA1 configuration has the best test performance according to the $\mu\Delta$ metric. ImageNet-based pre-training outperforms the RadImageNet one by a large margin. Nevertheless, the WSS DSC results on the challenging Cheng et al. dataset are generally lower when compared to fully supervised approaches e.g. [Sahoo et al., 2023] that reports a DSC of 0.9011. Quantitative results also show that meningioma distances $\Delta_m$ perform better (are consistently lower) than glioma $\Delta_g$

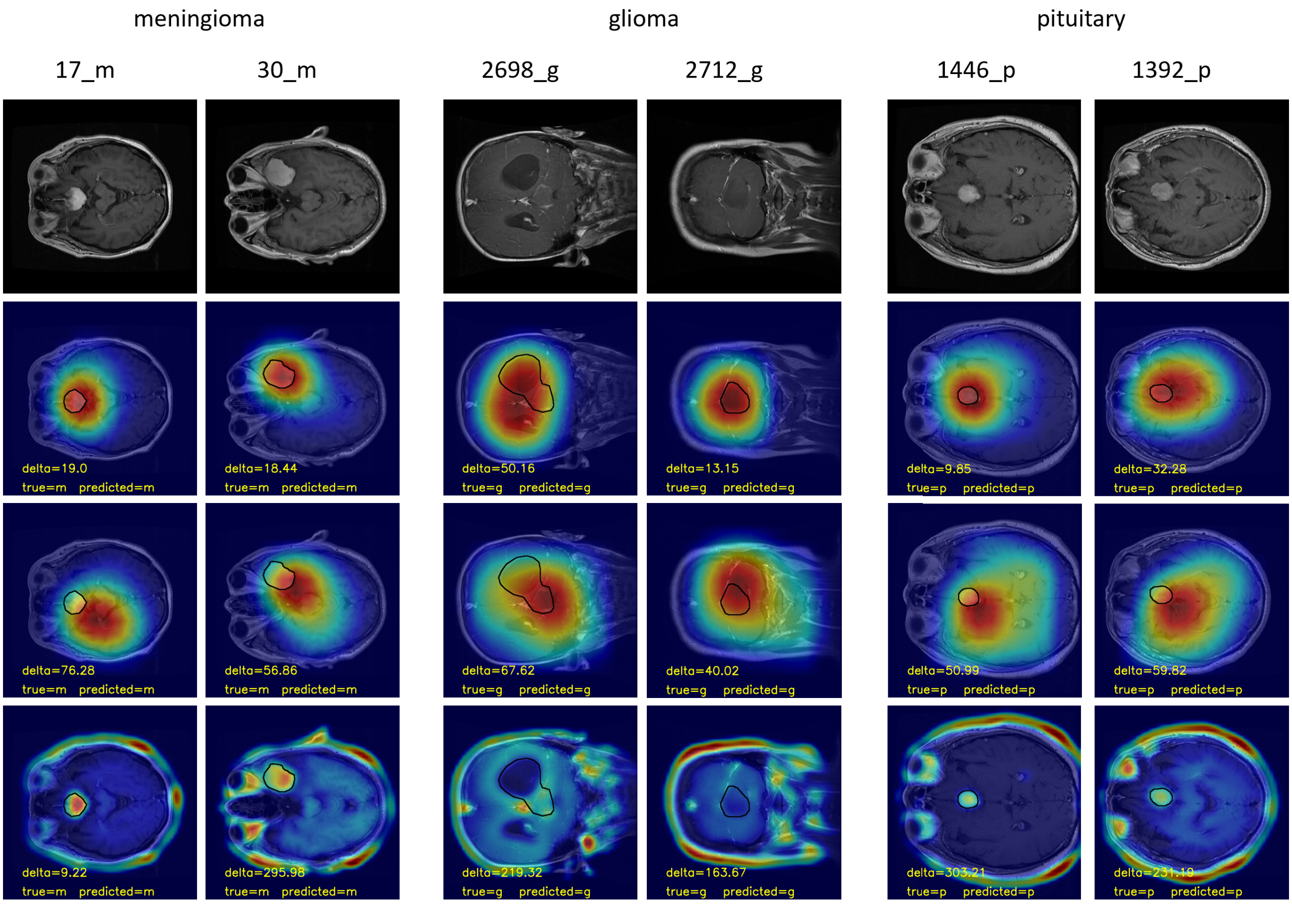


Figure 4: Sampled qualitative results are provided for each tumour class. In the EigenCAM localization heatmaps overlaid on the original images, the color red represents a higher level of pixel importance for the final prediction. Ground truth tumour contours are shown in black overlaid on heatmaps. **From top to bottom**: **First row** shows original images with the image index. **Second row** shows SEA1 ImageNet related heatmaps. **Third row** shows SEA3 RadImageNet related heatmaps. **Last row** shows heatmaps obtained by training the B2-Net from scratch.

and pituitary $\Delta_p$ distances.

Our qualitative results in Figure 4 show that despite varying tumour image characteristics (dark glioma tumours versus bright meningioma and pituitary tumours), our XAIMed-Net is able to focus on the appropriate tumour regions. For our qualitative comparison in Figure 4 we compare XAIMed-Net also with a shallow convolutional network which we term B2-Net. This model consists of three convolutional blocks. The first block is **Conv2d**(3,16,5,5) → **ReLU** → **MaxPool**(2,2). The second block is **Conv2d**(16,32,4,4) → **ReLU** → **MaxPool**(2,2). The last block is **Conv2d**(32,64,3,3) → **ReLU** → **MaxPool**(2,2). It is followed by a classifier with two fully connected layers **Flatten** → **fc1**(40000,16) → **ReLU** → **fc2**(16,3) → **softmax**. It can be seen from Figure 4 (last row) that the B2-Net has the least ability to focus on tumours. However, we did not perform skull stripping in our experiments. This pre-processing procedure could improve the results. It appears that the B2-Net model focuses rather on bright image areas. Figure 4 also shows that SEA1 ImageNet-based XAIMed-Net focuses on tumour areas better than the RadImageNet-based SEA3 model.

## 5 Discussion, Limitations and Future Directions

Interestingly, the XAIMed-Net pre-trained on RadImageNet shows worse WSS performance than the XAIMed-Net pre-trained on the ImageNet dataset, despite that pre-training on grayscale images can allow the training of more generalizable low-level filters in the initial layers of the network as claimed by the authors [Mei et al., 2022]. Through our experiments, this inferior performance of the XAIMed-Net pre-trained on RadImageNet is confirmed both quantitatively and qualitatively. However, we did not get the heatmaps validated by experienced radiologists and it is possible that RadImageNet-based XAIMed-Net focuses on more relevant anatomical areas. It does, however, focus less on tumours. Generally, RadImageNet and ImageNet pre-trained XAIMed-Net shows different training behaviours. In both cases, the spread of the heatmap is large and its shape is mostly round or elliptical, most likely due to the use of $3 \times 3$ convolutional kernels. As a result, large round tumours are better localized. Our future work will investigate the interconnection between squeeze and expansion factors and the Gaussian spread factors. The performance depends on channel compression and expansion rates. A question remains how to obtain optimal compression and expansion parameter values automatically.

While the quantitative metrics were computed on the whole test set, the above qualitative results, however, show only the selected *success* cases of XAIMed-Net and we will aim at understanding and addressing *failure* cases in our future work. Also, we will aim at testing our XAIMed-Net on other medical image classification datasets.

## 6 Conclusion

In this paper we developed an explainable classification network, XAIMed-Net, and presented a methodology to quantify the explainability of brain tumour classification of the Cheng et al. dataset. Although our WSS results are below what can be achieved using fully supervised image segmentation methods, the idea of XAIMed-Net looks promising. Going forward, we believe that we can achieve better results with tuning the reduction, compression and Gaussian spread parameters of the individual layers of the convolutional head in XAIMed-Net.

## Acknowledgments

This work was funded by Science Foundation Ireland through the SFI Centre for Research Training in Machine Learning (18/CRT/6183).

## References

[Ahmed et al., 2023] Ahmed, F., Asif, M., Saleem, M., Mushtaq, U. F., and Imran, M. (2023). Identification and prediction of brain tumor using VGG-16 empowered with explainable artificial intelligence. *International Journal of Computational and Innovative Sciences*, 2(2):24–33.

[Antoniadi et al., 2021] Antoniadi, A. M., Du, Y., Guendouz, Y., Wei, L., Mazo, C., Becker, B. A., and Mooney, C. (2021). Current challenges and future opportunities for xai in machine learning-based clinical decision support systems: A systematic review. *Applied Sciences*, 11(11).

[Bernal and Mazo, 2022] Bernal, J. and Mazo, C. (2022). Transparency of artificial intelligence in healthcare: Insights from professionals in computing and healthcare worldwide. *Applied Sciences*, 12(20).

[Chatterjee et al., 2023] Chatterjee, S., Yassin, H., Dubost, F., Nürnberger, A., and Speck, O. (2023). Weakly-supervised segmentation using inherently-explainable classification models and their application to brain tumour classification. *arXiv https://arxiv.org/abs/2206.05148*.

[Cheng et al., 2015] Cheng, J., Huang, W., Cao, S., Yang, R., Yang, W., Yun, Z., Wang, Z., and Feng, Q. (2015). Enhanced performance of brain tumor classification via tumor region augmentation and partition. *PLoS ONE*, 10(10: e0140381).

[Gaur et al., 2022] Gaur, L., Bhandari, M., Razdan, T., Mallik, S., and Zhao, Z. (2022). Explanation-driven deep learning model for prediction of brain tumour status using mri image data. *Frontiers in Genetics*, 13.

[Hamada, 2020] Hamada, A. (2020). Br35h :: Brain tumor detection 2020. *available on https://www.kaggle.com/datasets/ahmedhamada0/brain-tumor-detection*.

[He et al., 2015] He, K., Zhang, X., Ren, S., and Sun, J. (2015). Deep residual learning for image recognition. *arXiv https://arxiv.org/abs/1512.03385*.

[Iandola et al., 2017] Iandola, F. N., Han, S., Moskewicz, M. W., Ashraf, K., Dally, W. J., and Keutzer, K. (2017). Squeezenet: AlexNet-level accuracy with 50x fewer parameters and <0.5MB model size. *The International Conference on Learning Representations*.

[Kumar et al., 2021] Kumar, A., Manikandan, R., Kose, U., Gupta, D., and Satapathy, S. C. (2021). Doctor's dilemma: Evaluating an explainable subtractive spatial lightweight convolutional neural network for brain tumor diagnosis. *ACM Trans. Multimedia Comput. Commun. Appl.*, 17(3s).

[Loshchilov and Hutter, 2019] Loshchilov, I. and Hutter, F. (2019). Decoupled weight decay regularization. *The International Conference on Learning Representations*.

[Mei et al., 2022] Mei, X., Liu, Z., Robson, P. M., Marinelli, B., Huang, M., Doshi, A., Jacobi, A., Cao, C., Link, K. E., Yang, T., Wang, Y., Greenspan, H., Deyer, T., Fayad, Z. A., and Yang, Y. (2022). RadImageNet: An open radiologic deep learning research dataset for effective transfer learning. *Radiology: Artificial Intelligence*, 4(5).

[Muhammad and Yeasin, 2020] Muhammad, M. B. and Yeasin, M. (2020). Eigen-CAM: Class activation map using principal components. *International Joint Conference on Neural Networks (IJCNN)*.

[Ruan et al., 2021] Ruan, D., Wang, D., Zheng, Y., Zheng, N., and Zheng, M. (2021). Gaussian Context Transformer. *The IEEE Conference on Computer Vision and Pattern Recognition*.

[Russakovsky et al., 2015] Russakovsky, O., Deng, J., Su, H., Krause, J., Satheesh, S., Ma, S., Huang, Z., Karpathy, A., Khosla, A., Bernstein, M., Berg, A. C., and Fei-Fei, L. (2015). ImageNet large scale visual recognition challenge. *International Journal of Computer Vision*.

[Sahoo et al., 2023] Sahoo, A. K., Parida, P., Muralibabu, K., and Dash, S. (2023). Efficient simultaneous segmentation and classification of brain tumors from MRI scans using deep learning. *Biocybernetics and Biomedical Engineering*, 43(3):616–633.

[Saranya A., 2023] Saranya A., S. R. (2023). A systematic review of explainable artificial intelligence models and applications: Recent developments and future trends. *Decision Analytics Journal*, 7:100230.

[Schiavon et al., 2023] Schiavon, D. E. B., Becker, C. D. L., Botelho, V. R., and Pianoski, T. A. (2023). Interpreting convolutional neural networks for brain tumor classification: An explainable artificial intelligence approach. In Naldi, M. C. and Bianchi, R. A. C., editors, *Intelligent Systems*, pages 77–91, Cham. Springer Nature Switzerland.

[Sutton and McCallum, 2010] Sutton, C. and McCallum, A. (2010). An introduction to Conditional Random Fields. *arXiv https://arxiv.org/abs/1011.4088*.

[Zeineldin et al., 2022] Zeineldin, R. A., Karar, M. E., Elshaer, Z., Coburger, J., Wirtz, C. R., Burgert, O., and Mathis-Ullrich, F. (2022). Explainability of deep neural networks for mri analysis of brain tumors. *International Journal of Computer Assisted Radiology and Surgery*, 17:1673–1683.